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JAVA
LESSON 2

INTRODUCTION TO JAVA

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JAVA OVERVIEW

Java is a popular programming language, created in 1995.

It is owned by Oracle, and more than 3 billion devices run Java.

It is used for:

- Mobile applications (specially Android apps)
- Desktop applications
- Web applications
- Web servers and application servers
- Games
- Database connection
- And much, much more!

JAVA OVERVIEW

- Java works on different platforms (Windows, Mac, Linux, Raspberry Pi, etc.)
- It is one of the most popular programming languages in the world
- It has a large demand in the current job market
- It is easy to learn and simple to use
- It is open-source and free
- It is secure, fast and powerful
- It has huge community support (tens of millions of developers)
- Java is an object oriented language which gives a clear structure to programs and allows code to be reused, lowering development costs
- As Java is close to C++ and C#, it makes it easy for programmers to switch to Java or vice versa

JAVA OVERVIEW

JVM (Java Virtual Machine): An abstract machine that enables your computer to run Java programs. It converts Java bytecode into machine code.

Platform Independence: Java code is written once and can be run anywhere without modification. This is achieved through the use of bytecode and the JVM.

JRE (Java Runtime Environment): Includes the JVM and the necessary libraries and components to run Java applications.

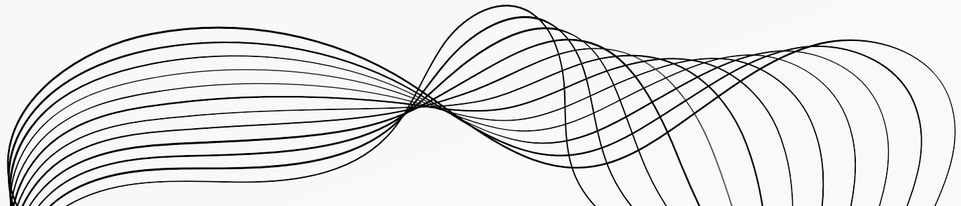
JDK (Java Development Kit): A complete toolkit that includes the JRE and development tools such as the Java compiler (javac).

JAVA LANGUAGE SYNTAX AND STRUCTURE

class

```
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
    }
}
```

main method



JAVA LANGUAGE SYNTAX AND STRUCTURE

```java
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
    }
}
```

public: Accessible from anywhere.
static: Belongs to the class rather than any instance of the class.
void: Does not return any value.
main: Name of the method.
String[] args: Parameter to the main method, an array of strings.

JAVA LANGUAGE SYNTAX AND STRUCTURE

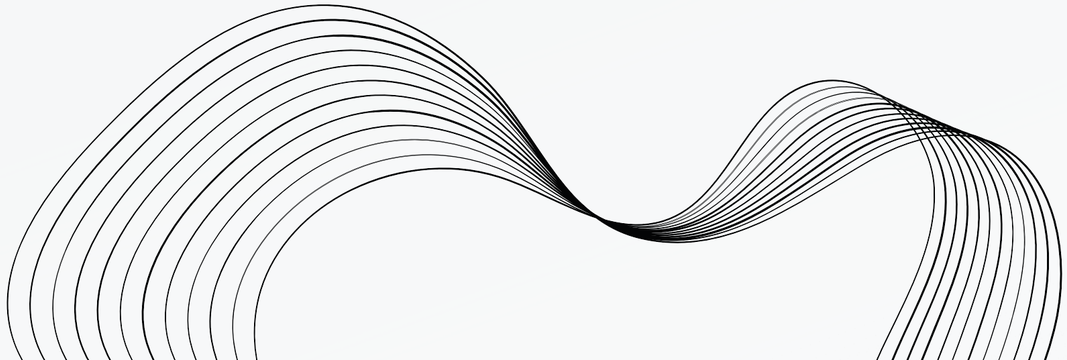
```
// This is a single-line comment
```

```
/* This is a
   multi-line comment */
```

```
/**
 * This is a documentation comment.
 * @param args Command line arguments
 */
```

PRACTICE

1. Write a simple program to print "Hello, World!".
2. Modify the program to include multi-line and single-line comments.
3. Write a program that prints a welcome message with your name.



NEXT LESSON

DATA TYPES, VARIABLES, AND OPERATORS